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Enhancing anaerobic digestion of swine manure using microbial electrolysis cell and microaeration

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ABSTRACT

To enhance anaerobic digestion (AD) of swine manure (SW) and decrease the hydrogen sulfide (H_2S) concentration in biogas, microbial electrolysis cell (MEC) and microaeration (MA) were applied, and semi-continuous experiments were conducted to compare their enhancements. The results showed that both MEC and MA enhanced the production of methane (24.74–68.24 %) and the removal of organic matter (8.54–23.80 %), increased the reaction rates of the four main AD steps (hydrolysis, acidogenesis, acetogenesis, and methanogenesis), and boosted the activities of key enzymes, with MEC showing the greater increase. A slight synergistic effect was observed between MEC and MA. MA and MA combined with MEC (MM) decreased the H_2S concentration in biogas (67.93 ± 14.20 % and 12.94 ± 17.81 %) by enriching sulfur oxidizing bacteria, boosting sulfite oxidase activity, and restricting sulfite reductase activity. However, MEC increased the H_2S concentration in biogas (175.26 ± 96.80 %) by enriching sulfate reducing bacteria and boosting the sulfite reductase activity. The three methods (MEC, MA, and MM) all enriched the hydrolytic and acidogenic bacteria and altered the predominant methanogenesis pathway. MEC and MM strengthened direct interspecies electron transfer by enriching *Methanosaeta, Clostridium_sensu_stricto_1*, and *Romboutsia*. Results from this study may offer some promising alternatives for the energy issues faced by villages and small towns in China.

1. Introduction

China is the largest swine producer and swine meat consumer in the world. The huge demand for meat and rapid expansion of the swine farming industry have led to the large-scale generation of swine manure (SM) in China [1], which has reached approximately 0.6 billion tons annually [2,3]. Owing to its high chemical oxygen demand (COD) and nitrogen and phosphorus content, SM causes serious environmental problems (such as eutrophication, odor, and pathogen transmission) and has become an important agricultural source of pollution in villages and small towns in China [2,3]. For example, SM can cause eutrophication of fresh water resources due to its high content of nitrogen (N) and phosphates (P), while the emission of odors and volatile compounds from SM can significantly impact human health due to the high ammonia and volatile compounds content [4]. However, SM is also regarded as a resource, with farmers applying it as a fertilizer to enhance crop production for centuries [4]. Therefore, the treatment of SM has become a challenge for maximum productivity with minimum environmental impact and appropriate treatment of SM is important.

Anaerobic digestion (AD) is considered a promising and environmentally sustainable technology for SM treatment because it can convert organic matter into bioenergy (methane and hydrogen), reduce the amount of SM, and generate organic fertilizers [5,6]. In addition, the AD of SM can provide bioenergy for villages and small towns in China, where swine farming is predominantly located. However, the AD of SM has certain shortcomings, such as slow hydrolysis-acidification rate, low methane yield, insufficient substrate utilization, and high amounts of solid and liquid digestates [5,6]. Previous studies have shown that the methane yield of SM ranged from 220 to 438 mL/g volatile solids (VS), which is far lower than its theoretical value (720 mL/g VS) [7,8]. Moreover, there is also a problem of hydrogen sulfide (H_2S) production during the AD of SM, which inhibits methanogens and corrodes equipment and pipelines [9,10]. A previous study [11] showed that the sulfur content of SM is 5.56–5.99 g/kg total solids (TS). Therefore, to produce

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more high-quality bioenergy for villages and small towns in China, the AD of SM must be enhanced. Many methods and strategies have been developed to enhance this process, including pretreatment, co-digestion with high C/N organic matter, thermophilic digestion, adding additives, microbial electrolysis cell (MEC), and microaeration (MA) [12–14].

MEC is driven by electrogenic microorganisms under a low applied external voltage and considered a promising method for enhancing AD [15,16]. This environmentally friendly bio-electrochemical technology was first applied to produce biohydrogen from various organic wastes [6,15]. Recently, MEC has been used to enhance the AD of organic wastes by promoting electron transfer, enriching functional microorganisms, and changing the methanogenesis pathways [15,17]. MEC can increase electron transfer by strengthening the direct interspecies electron transfer (DIET) pathway, which is considered a promising strategy to improve the performance of AD [16]. Previous studies [14,18,19] have reported that MEC can improve the AD of SM. For example, Zou et al. [14] found that MEC with an applied voltage of 0.9 V increased the maximum cumulative biogas yields from 401.7 mL/g TS to 547.3 mL/g TS and the methane yield from 347.7 mL/g TS to 224.0 mL/g TS in batch AD of SM.

MA, namely, dosing a small amount of oxygen or air to the AD system, is another promising method for enhancing AD [20,21]. MA has recently been considered an attractive and effective strategy to enhance AD. MA-based AD has been defined as a micro-aerobic condition with an oxidation–reduction potential (ORP) ranging from −300 to 0 mV, as determined by a standard hydrogen electrode [21]. Micro-oxygen dosing enriches the facultative acidogens and increases the bacterial diversity during AD [21]. Moreover, it can promote the oxidation of sulfide, which decreases the concentration of H_2S in biogas [22]. Therefore, MA can enhance AD by accelerating hydrolysis and acidification, scavenging H_2S and reducing its concentration in biogas, improving process stabilization, regulating microbial communities, and promoting methane production through the supply of a small amount of oxygen [21–23]. Previous studies have found that MA can enhance the hydrolysis and acidification of SM [24] and promote organic matter removal and methane production rate in swine wastewater treatment [12].

Although MEC and MA present different mechanisms for enhancing AD, previous studies have not compared their enhancement effects. Moreover, considering the special characteristics of SM (such as high organic content and containing sulfur compounds), it is expected that MEC and MA can synergistically enhance the AD of SM by exerting their respective advantages when they are co-integrated. However, no study has yet addressed this topic. Therefore, to fill this gap, this study innovatively integrates MEC and MA and compares their individual and synergistic effects on the AD of SM. The performances (biogas production and quality, matter conversion, and digestate characteristics), reaction rates of the four AD steps, functional enzyme activities, and microbial community structures of the AD processes were evaluated. The objective is to explore the synergistic enhancements and mechanisms of MEC and MA and identify an effective method of boosting the AD of SM to help resolve the energy issues in villages and small towns in China.

2. Materials and methods

2.1. Swine manure and inoculum

The SM used in this study was obtained from a pig farm in Beijing, China. After collection, the SM was ground in a blender (QH-400D, Qihui, China) and then manually screened through a 40 mesh sieve to remove large particles. The pretreated SM was stored at 4 °C until use. The digestate used as the inoculum in the experiments was obtained from a lab-scale mesophilic anaerobic digester, which used SM as a substrate. The main characteristics of the SM and inoculum are summarized in Table 1.

Table 1
Main characteristics of swine manure and inoculum.

Item	Swine manure	Inoculum
pH	7.37 ± 0.11	7.96 ± 0.13
TS (g/L)	78.23 ± 3.87	38.34 ± 2.55
VS (g/L)	63.83 ± 3.26	27.52 ± 3.11
TCOD (g/L)	86.34 ± 1.57	31.34 ± 4.17
SCOD (g/L)	18.22 ± 0.37	2.36 ± 0.76
NH_4^+-N (g/L)	3.08 ± 0.33	0.71 ± 0.28
VFA (g/L)	8.91 ± 0.81	0.05 ± 0.01

2.2. Anaerobic digesters

Four identical lab-scale continuous stirred tank reactors (CSTRs) with total and working volumes of 5 L and 3.9 L, respectively, were used as anaerobic digesters. Two electrodes were integrated into each digester with a distance of 2 cm. The electrodes were composed of a carbon rod (diameter × height = 8 × 350 mm) and ten square carbon felts (length × width × thickness = 60 × 30 × 5 mm), which is similar to those used in previous study [17] (Fig. S1).

2.3. Operation of the anaerobic digesters

The four ADs were first inoculated with 3.9 L of inoculum and purged with high-purity nitrogen (99.99 %) for 5 min to eliminate oxygen and create anaerobic conditions. One digester (MA-AD) was continuously aerated with a micro-aerator (BT100-2 J, Longer Co., China) at an aeration rate of 0.014 L/L/d (based on the working volume) [25,26]. The second digester (MEC-AD) was operated at 1.2 V applied voltage through a DC power supply [17,27]. The third digester (MM-AD) was simultaneously subjected to 1.2 V voltage and continuously aerated at a rate of 0.014 L/L/d. The last digester (Con-AD), which was not connected to the DC power supply applied voltage and not aerated, served as the control. The four ADs were maintained at 37 ± 1°C using heaters (CH-1006, Hengping Co., China) and water jackets, and mixed at a stirring rate of 300 rpm. Subsequently, the digesters were started up in a semi-continuous mode with daily discharge and feeding using gastight glass syringes. The hydraulic retention times (HRTs) of the four ADs were decreased from 100 d to 30 d over 2 weeks by gradually increasing the feed, and then the digesters were operated for 88 d at an HRT of 30 d. The four ADs were operated for 104 d in total, with the operation period dividing into start-up (0–30 d), adjustment (31–60 d), and stable (61–104 d) stages.

2.4. Analytical methods

The biogas produced from the four ADs was stored in gas bags, and its volume was measured every two days by replacing saturated NaCl aqueous solution [6]. The methane (CH_4) content and H_2S concentration in the biogas were measured using two gas chromatographs (GCs): one equipped with a flame ionization detector (Agilent 6890 N, USA) for CH_4, and the other with a flame photometric detector (Agilent 6890, USA) for H_2S [17].

The influents (substrate, SM) and effluents (anaerobic digestates) of the four ADs were sampled and analyzed twice a week. The analytical parameters included total solids (TS), volatile solids (VS), chemical oxygen demand (COD), pH, ORP, total ammonia nitrogen (TAN), total alkalinity (TA), and volatile fatty acids (VFAs). The analytical methods were the same as those used in previous studies [6,17] and the ORP was measured using a multi-parameter portable meter (WTW Multi 3630 IDS, Langenhagen, Germany). The total sulfur content was measured using an elemental analyzer (CHWKL-0253, Shanghai Changai, China). Sulfate (SO_4^{2-}) was measured using ion chromatography (ICS3000, Dionex, Sunnyvale, CA, USA); sulfide (S^{2-}) was measured using a gas-phase molecular absorption spectrometer (GMA3376, Shanghai Beiyu, China); and elemental sulfur (S^0) was measured using an infrared carbon sulfur

analyzer (HX-HW8B, Nanjing Huaxin, China). The activities of α-glucosidase, protease, dehydrogenase (DHA), acetate kinase, sulfite reductase, and sulfite oxidase in the digestates from the four ADs were measured using ELISA kits (Shanghai Enzyme-linked Biotechnology Co., Ltd., China) [28]. Coenzyme F420 activity was measured using the method described by Shao et al. [29]. The relative activities of the above enzymes were used in this study, with the enzyme activities of samples from the Con-AD being defined as 100 %.

The digestates from the four ADs were sampled at the end of the stable stage (102 d) for microbial communities analysis based on previously reported methods [6]. The detailed process is as follows. First, the total genomic DNA of the sampled digestates was extracted using Omega Mag-Bind soil DNA kits (Omega Bio-Tek, Norcross, GA, USA). The qualified DNAs were used as templates for polymerase chain reaction (PCR) amplification with the bacterial primers of 338F (5′-ACTCCTACGGGAGGCAGCAG-3′) and 806R (5′-GGACTACHVGGGTWTCTAAT-3′) for the V3–V4 hypervariable region and archaeal primers of 524F10extF (5′-TGYCAGCCGCCGCGGTAA-3′) and Arch958RmodR (5′-YCCGGCGTTGAVTCCAATT-3′) for 16S rRNA. The PCR conditions were similar to those used in a previous study [6]. The amplicons were extracted from 2 % agarose gels, purified using an AxyPrep DNA Gel Extraction Kit (Axygen Biosciences, Union City, CA, USA) according to the manufacturer's instructions. Purified amplicons were pooled in equimolar amounts and paired-end sequenced (2 × 300) on an Illumina MiSeq platform (Illumina, San Diego, USA) following the standard protocols of Majorbio Bio-Pharm Technology Co. Ltd. (Shanghai, China). The sequencing data were processed in a similar manner to that used in a previous study [6].

2.5. Calculation and statistical analysis

The reaction rates of the four main AD steps (hydrolysis, acidogenesis, acetogenesis, and methanogenesis), mass balances based on COD, and sulfur balances for the four ADs were calculated based on previously reported methods [30–32]. The detailed calculations for the mass (Eq.1–5) and sulfur (Eq.6–11) balances for the ADs are provided as follows.

$$C_{SCOD}\ (\%) = \frac{SCOD}{TCOD_{Sub.}} \times 100 \quad (1)$$

$$C_{PCOD}\ (\%) = \frac{PCOD}{TCOD_{Sub.}} \times 100 \quad (2)$$

$$C_{CH4-COD}\ (\%) = \frac{CH_4 - COD}{TCOD_{Sub.}} \times 100 \quad (3)$$

$$PCOD = TCOD - SCOD \quad (4)$$

$$CH_4 - COD = \frac{Y_{CH4}}{0.35 \times V_{Sub.}} \quad (5)$$

$$C_{Sulfide}\ (\%) = \frac{\text{Sulfide}}{\text{total sulfur}} \times 100 \quad (6)$$

$$C_{Sulfate}\ (\%) = \frac{Sulfate}{totalsulfur} \times 100 \quad (7)$$

$$C_{Sulfur}\ (\%) = \frac{Sulfur}{totalsulfur} \times 100 \quad (8)$$

$$C_{H_2S}\ (\%) = \frac{H_2S}{totalsulfur} \times 100 \quad (9)$$

$$C_{Other}\ (\%) = (1 - C_{Sulfide} - C_{Sulfate} - C_{Sulfur} - C_{H_2S}) \times 100 \quad (10)$$

$$H_2S = \frac{34}{22.4} \times 3.9 \times C_{H_2S} \times Y_{Biogas} \quad (11)$$

where, C_{SCOD} is the weight content of SCOD (%); SCOD is the SCOD of substrate or digestate (g/L); $TCOD_{sub.}$ is the TCOD of substrate (g/L); C_{PCOD} is the weight content of particle COD (PCOD) (%); PCOD is the SCOD of substrate or digestate (g/L); $C_{CH4\text{-}COD}$ is the weight content of CH_4-COD (%); CH_4-COD is the CH_4 yield presented as COD (g/L); Y_{CH4} is the CH_4 yield (mL/d); 0.35 is the theoretical CH_4 yield of COD (mL/g COD); $V_{Sub.}$ is the influent flow of substrate (L/d); $C_{Sulfide}$ is the weight content of sulfide (%); Sulfide is the sulfide concentration in substrate or digestate (g/L); $C_{Sulfate}$ is the weight content of sulfate (%); Sulfate is the sulfate concentration in substrate or digestate (g/L); C_{Sulfur} is the weight content of sulfur(%); Sulfur is the sulfur concentration in substrate or digestate (g/L); Total sulfur is the total sulfur concentration in substrate (g/L); C_{H2S} is the H_2S concentration in biogas (L/L); Y_{Biogas} is the biogas production(ml/L/d); 34 is the molecular weight H_2S; 22.4 is the molar volume of gas (L/mol); and 3.9 is the working volume of anaerobic digester (L).

Analysis of variance (ANOVA) was performed to identify statistically significant differences, and Tukey's honestly significant difference (HSD) test was performed to separate the means using SPSS 22.0 (USA). The results were considered significant at the 95 % confidence level ($p < 0.05$) [33]. The correlations among microbial communities, main environmental factors, and biogas production were analyzed using redundancy analysis (RDA), which was conducted using the CANOCO 4.5 package [6].

3. Results and discussion

3.1. Biogas production

Biogas production of the four ADs was measured, and the results were summarized in Fig. 1 and Table 2. The results showed that biogas production of all four ADs gradually increased during the start-up stage (0–30 d), fluctuated during the adjustment stage (31–60 d), and then entered the stable stage, with averages of 0.58 ± 0.03 L/L/d (Con-AD), 0.75 ± 0.04 L/L/d (MA-AD), 0.92 ± 0.03 L/L/d (MEC-AD), and 0.96 ± 0.02 L/L/d (MM-AD) (Fig. 1-a and Table 2). The methane contents in the biogas produced from the four ADs were relatively stable during the adjustment and stable stages and ranged from 52.57 % to 79.67 %. The methane contents of MEC-AD and MM-AD (63.17 ± 0.98 % and 61.86 ± 1.27 %, respectively) were higher than those of Con-AD (61.11 ± 2.11 %) and MA-AD, which had the lowest content (59.20 ± 1.42 %) (Fig. 1-b and Table 2). Statistical analysis showed that the methane contents in the biogas produced from Con-AD and MM-AD were similar ($p = 0.12 > 0.05$, Table S1) and significantly different from those of the other two ADs ($p < 0.05$, Table S1).

Methane production of the four ADs was obtained from biogas production and methane content (Fig. 1-c). The changes in methane production were similar to those in biogas production, and the methane production of Con-AD was the lowest (Fig. 1-c and Table 2). The results indicated that MEC and MA enhanced the production of biogas and methane at rates of 28.21–65.97 % and 24.74–68.24 %, respectively (Table S2) [15,21], and MEC presented a greater enhancing effect than MA. Synergism was also observed when MEC and MA were co-integrated during the AD of SM (Table S2). The coefficients of variation (Table S3) showed that the three methods (MEC, MA, and MM) decreased the variations in methane content and methane production. Therefore, MEC and MA enhanced the stability of methane production [15,21]. In addition, the three methods presented greater enhancing effects on the AD of SM relative to those reported in most previous studies (Table 3). For example, Liu et al. [5] found that MEC enhanced the methane yield of SM by 60.89 % and de Oliveira et al. [12] found that MA enhanced the methane yield of swine wastewater by 57.54 %.

The H_2S concentrations in biogas produced from the four ADs were also measured (Fig. 1-d). The results showed that the H_2S concentrations in biogas produced from MEC-AD (with an average of 3424 ± 770

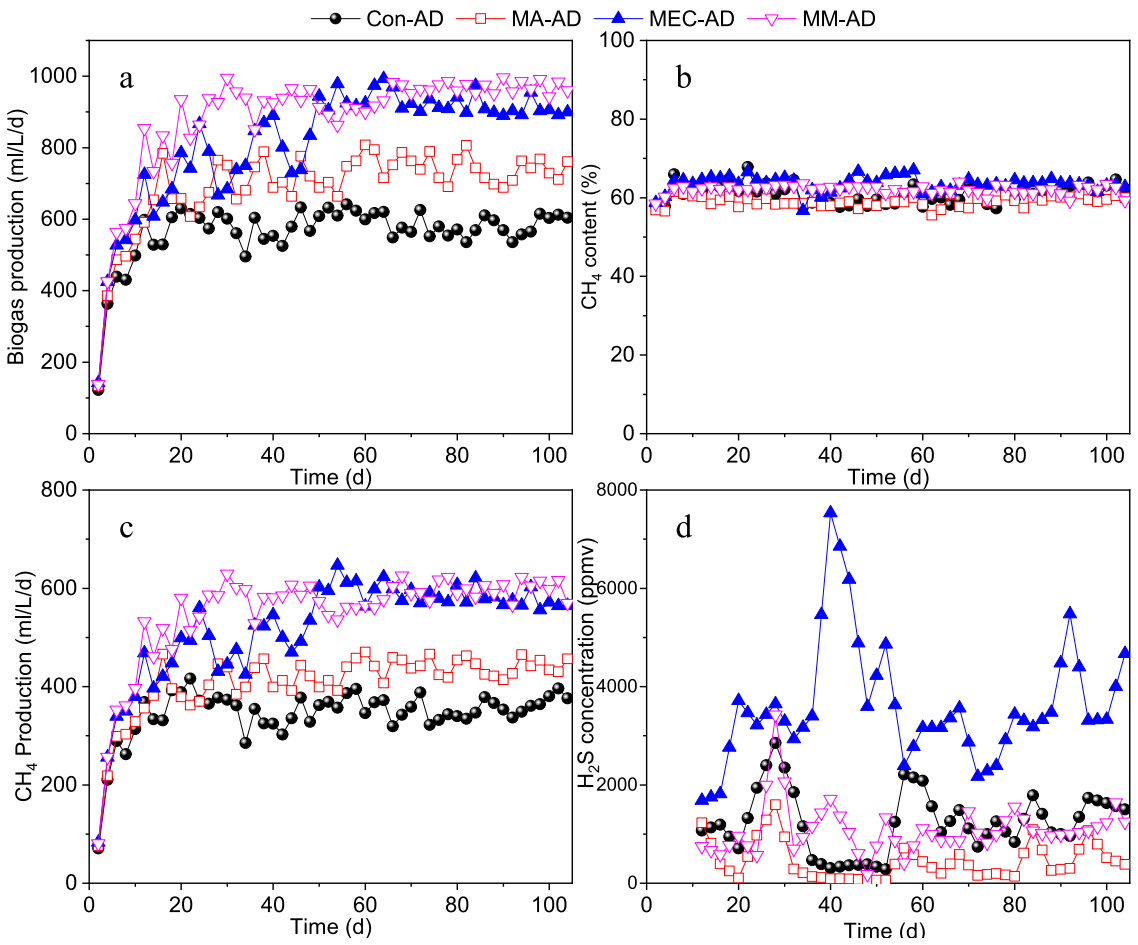


Fig. 1. Biogas production of the four ADs during the experiment.

Table 2
Average biogas production of the four ADs during the stable period.

Item	Con-AD	MA-AD	MEC-AD	MM-AD
Biogas production (mL/L/d)	582 ± 28	746 ± 37	922 ± 31	963 ± 24
CH_4 content (%)	61.06 ± 2.04	59.20 ± 1.42	63.17 ± 0.98	61.86 ± 1.27
H_2S concentration (ppmV)	1321 ± 342	446 ± 273	3424 ± 770	1112 ± 219
CH_4 production (mL/L/d)	355 ± 20	441 ± 18	582 ± 19	596 ± 20
H_2S production (mL/L/d)	0.77 ± 0.01	0.33 ± 0.01	3.16 ± 0.02	1.07 ± 0.01
Biogas yield (mL/g In.* VS)	273.41 ± 13.32	350.55 ± 28.71	433.39 ± 25.80	452.55 ± 20.91
CH_4 yield (mL/g In. VS)	166.93 ± 9.52	207.40 ± 14.24	273.72 ± 15.79	279.94 ± 17.59

* In.: Influent.

Table 3
Comparison of the results from this work and previous studies.

Enhancement method	Methane yield enhancement (%)	Experimental mode	Reference
Thermal steam explosion pretreatment	106.92	Batch	[34]
Zero-valent iron	54.18	Batch	[35]
Ferric oxide	11.06	Batch	[36]
Zero-valent iron	36.9	Batch	[37]
Nano-bubble water	19–39	Batch	[38]
Ferrous chloride	21.5	Batch	[39]
MEC	60.89	Batch	[5]
Biochar	25	Batch	[40]
MEC	18.85–53.17	Batch	[14]
MA	57.54	Continuous	[12]
Continuous NH_3 extraction	49	Continuous	[41]
MA	24.24 ± 5.68	Continuous	This study
MEC	63.97 ± 6.21	Continuous	This study
MM	67.70 ± 6.13	Continuous	This study

ppmV) were far higher than those from the other three digesters (Fig. 1-d and Table 2), which indicated that MEC enhanced the anaerobic conversion of sulfur-containing compounds (such as methionine, cysteine, sulfate (SO_4^{2-}) and sulfite (SO_3^{2-}) in the SM [42]. The H_2S concentrations in biogas produced from MA-AD and MM-AD (446 ± 273 and 1112 ± 219 ppmV, respectively) were lower than that from Con-AD (1321 ± 342 ppmV) because oxygen promoted the oxidation of sulfide [22]. The results suggested that the co-integration of MA and MEC for the AD (MM-AD) of SM improved the quality of biogas by increasing the methane content and decreasing the H_2S concentrations. Previous studies [10,43] showed that H_2S can decrease the methane production and even lead to process failure by inhibiting methanogens with an inhibitory concentration of 50–800 mg/L (about 0.33–5.27 × 10^5 ppmV). Because the H_2S concentrations in biogas produced in the four ADs were far lower than its inhibitory concentration, the effect of H_2S on the biogas production of the four ADs could be ignored. Additionally, the reduction of H_2S concentration by MA and MM may reduce equipment

corrosion.

3.2. Changes in the liquid phase

The digestate characteristics were detected during the experiment, and their changes are shown in Fig. 2. During the stable stage, the ORPs of MA-AD and MM-AD (−228 to − 286 mV and − 289 to − 300 mV, respectively) were higher than that of Con-AD and MEC-AD (−316 to − 328 mV and − 331 to − 340 mV, respectively) (Fig. 2-A and Table S4). These results indicated that the former two digesters were under micro-aerobic conditions and the latter two were under anaerobic conditions [21]. The pH of the digestates from the four ADs ranged from 7.68 to 8.15, which was similar to previous studies [17,32]. The TAN concentrations in the digestates from the four ADs gradually increased during the start-up and adjustment stages and reached a relatively steady level during the stable stage (Fig. 2-c). During the stable stage, the TAN concentration in digestate from Con-AD was the lowest (with an average of 3.66 ± 0.32 g/L), while that from MM-AD was the highest (with an average of 4.19 ± 0.28 g/L) (Fig. 2-c and Table S4). The results indicated that MA and MEC promoted the anaerobic conversion of nitrogen-containing substances (mainly proteins) and presented synergistic effects since TAN is a product of the anaerobic conversion of nitrogen-containing substances [44]. TAN toxicity is a known problem for AD and the reported TAN inhibitory threshold ranged between 1.50–7.00 g/L [45]. Therefore, TAN inhibition may occur in the four ADs. Free ammonia (FAN) is the active component that causes ammonia inhibition which depends mainly on TAN, temperature, and pH. During the stable stage, the FAN concentration in digestate from MEC-AD was the lowest (0.90 ± 0.00 g/L), while that from MM-AD was the highest (1.06 ± 0.00 g/L) (Table S4) [32]. The results indicated that the inhibition of TAN in the MM-AD was the highest and that in MEC-AD was the lowest. In other words, the three methods affected the TAN inhibition in the AD of SM.

The total VFAs (TVFAs) concentration in digestates from the four ADs ranged from 0.03 to 0.43 mg/L, which were far lower than that of the SM (7.04 mg/L to 9.73 mg/L), indicating that VFA accumulation did not occur during the experiment [46,47]. The low TVFAs concentrations in digestates also suggested that the TAN inhibition on methanogens was low. The digestates from the four ADs had high TA concentrations (9.01–15.51 g $CaCO_3$/L), indicating that the digestates had a good buffer ability [46]. The TVFA/TA ratios for the four ADs during the experiment were all lower than 0.3 (Fig. 2-f), indicating that the four ADs had stable operations [17]. The TS, VS, and total chemical oxygen demand (TCOD) concentrations in digestates from the four ADs during the experiment were also relatively stable, with Con-AD and MM-AD presenting the highest and lowest average values, respectively (Fig. 2-g-i and Table S4). Additionally, statistical analysis indicated that the TS, VS, and TCOD concentrations in digestates from the four ADs were significantly different ($p < 0.05$; Table S1). The results suggested that the three methods enhanced the removal of organic matter at an enhancement rate of 8.54–23.80 % (Fig. S2), which was consistent with the production of methane [46].

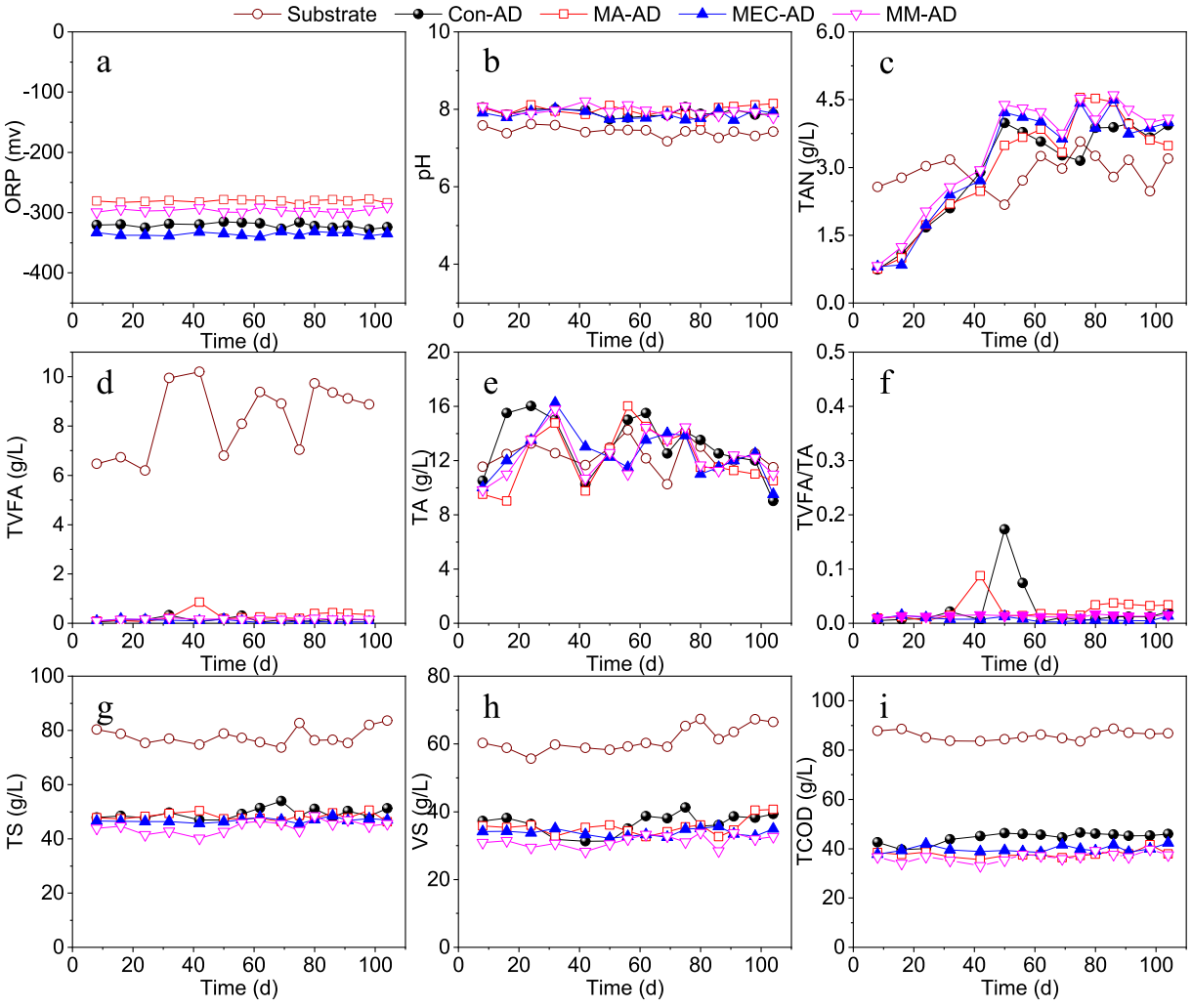


Fig. 2. Changes in characteristics of SM and digestates in the experiment.

3.3. Mass balance based on COD and sulfur

The mass balances based on COD for the four ADs during the stable stage are summarized in Fig. 3-a. Particle COD (PCOD) is an important component of SM and the digestates from the four ADs and accounted for 39.40–78.90 % of the COD, which was far higher than that of soluble COD (SCOD) (4.09–21.10 %). The content of CH_4-COD in MM-AD (59.13 ± 3.16 %) was the highest, followed by that in MEC-AD (57.82 ± 2.51 %) and Con-AD (35.26 ± 1.25 %), which was consistent with their CH_4 production (Fig. 1 and Table 2). The COD-gap value was also the highest (11.87 ± 0.41 %) in Con-AD and the lowest (−2.62 ± 0.09 %) in MM-AD. The results indicated that all three methods decreased the COD-gap. In other words, both MEC and MA decreased the loss of COD and the invalid anaerobic conversion of organic matter, exhibiting a slight synergistic effect [17].

Among the four ADs, the total sulfur concentration in digestates from MA-AD (10.45 ± 0.56 mg/g TS) was the highest, while that from MEC-AD (8.25 ± 0.19 mg/g TS) was the lowest (Table S4), which was consistent with their production of H_2S (Fig. 1-d). Meanwhile, the total sulfur concentrations in digestates were all higher than that in SM (7.05 ± 0.53 mg/g TS) due to the reduction of TS. Sulfate-S (SO_4^{2-}) was the main component of total sulfur in the SM, with a value of 86.81 ± 2.80 % (Fig. 3-b), while sulfide (S^{2-}) was the main component of total sulfur in the digestates from the four ADs, with a value of 40.24–57.49 % (Fig. 3-b). Among the sulfur balances, there were gaps between the four ADs (−3.36–6.25 %), which may be due to the deposition of sulfur at the bottom of the digesters or suspension of sulfur on the surface of the digestates [48]. This finding was based on the anaerobic conversion of sulfur-containing compounds into sulfide [48]. The elemental sulfur (S^0) content in the digestates from MA-AD and MM-AD (22.90 ± 0.81 % and 16.17 ± 1.02 %, respectively) were higher than that from Con-AD (4.54 ± 0.24 %) because H_2S was oxidized as shown in Eq. (1) owing to the injection of oxygen [21,22]. However, the S^0 content in the digestates from MEC-AD was the lowest among the four ADs (3.02 ± 0.16 %) (Fig. 3-b) because some sulfur was converted into H_2S and S^{2-} [21,22]. The H_2S content of MEC-AD was the highest (26.07 ± 0.89 %) and that of MA-AD was the lowest (2.75 ± 0.07 %) among the four ADs owing to their differences in H_2S production (Fig. 1-d and Table 2). Additionally, the content of others (e.g., organic sulfur) in Con-AD was the highest (18.17 ± 0.71 %) and that in MM-AD was the lowest (7.59 ± 0.25 %) (Fig. 3-b), indicating that MEC and MA also enhanced anaerobic conversion of this part of sulfur with MEC being stronger.

$$2HS^- + O_2 \rightarrow 2S^0 + 2OH^- \ (\Delta G^0 = -169.35\,KJ/mol) \quad (12)$$

Although sulfide can be oxidized into elemental sulfur (S^0), thiosulfate ($S_2O_3^{2-}$), or even sulfate (SO_4^{2-}) in a microaerobic environment [21], the sulfur balances showed that the sulfate contents in MA-AD and MM-AD were similar to that in Con-AD while those of elemental sulfur in MA-AD and MM-AD were far higher than that in Con-AD (Fig. 3-b). This finding indicated that sulfide was mostly oxidized into elemental sulfur (S^0). Therefore, the sulfur conversion pathways in the four ADs were slightly different (Fig. S3), with MEC and MM enhancing the conversion of other sulfur compounds to H_2S and S^{2-}; MA and MM enhancing the conversion of H_2S and S^{2-} to S^0; and S^0 potentially oxidizing to $S_2O_3^{2-}$ and SO_4^{2-} in the MA-AD and MM-AD [48].

3.4. Enzyme activity and reaction rates

The activities of typical hydrolysis-related enzymes (α-glucosaccharase and protease), acidogenesis (DHA)-related enzymes, acetogenesis-related enzymes (acetate kinase), methanogenesis-related

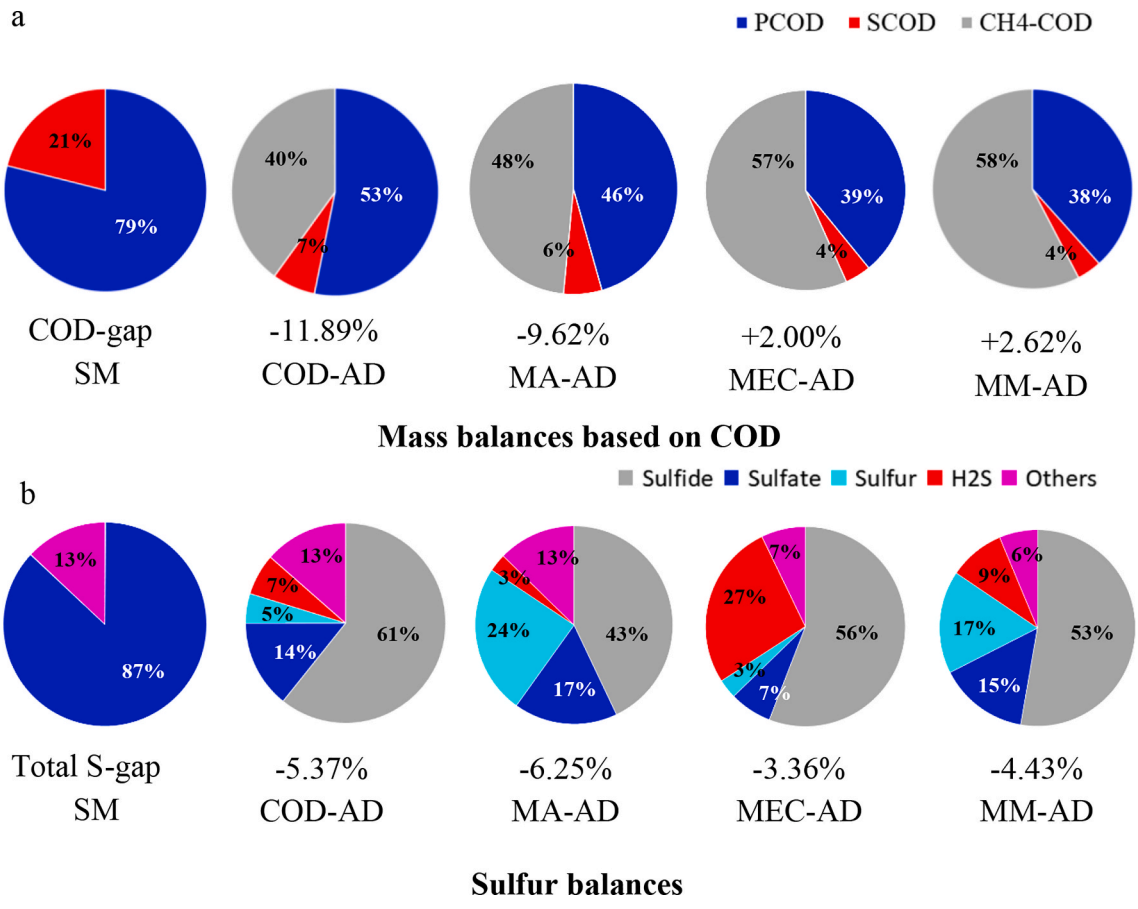


Fig. 3. Sulfur balance for the SM and four ADs during the stable stage.

enzymes (coenzyme F420), and sulfur conversion-related enzymes (sulfite reductase and sulfite oxidase) were assayed during the stable period (Fig. 4-a). The activities of all assayed enzymes in Con-AD were the lowest. The other three ADs had 118–142 % higher values than those in Con-AD, with MM-AD showing the highest activities, indicating that the three methods enhanced the activities of these enzymes. The activities of α-glucosaccharase and protease in MA-AD were higher than those in MEC-AD, while those of DHA and coenzyme F420 showed the opposite results. These results suggested that MA had greater enhancing effects on hydrolysis- and acidogenesis-related enzymes than MEC, while the opposite results were obtained for acetogenesis- and methanogenesis-related enzymes [28]. The activity of sulfite reductase in MEC-AD, MA-AD, and MM-AD was 112 ± 4 %, 89 ± 4 %, and 78 ± 5 % higher than that in Con-AD, respectively. However, the activity of sulfite oxidase was similar between MEC-AD and Con-AD, while that in MA-AD and MM-AD were higher than that in Con-AD, with that in MA-AD presenting the highest (Fig. 4-a). The results indicated that MEC enhanced the activity of sulfite reductase and did not affect the activity of sulfite oxidase, whereas MA and MM decreased the activity of sulfite reductase and increased the activity of sulfite oxidase [22]. The different activities of sulfite reductase and oxidase in the four ADs were consistent with the changes in sulfur (Fig. 3-b).

Previous studies [15,21] showed that both MA and MEC could accelerate the hydrolysis and acidogenesis of organic matter. Therefore, the reaction rates of hydrolysis, acidogenesis, acetogenesis, and methanogenesis (four main AD steps) were calculated (Fig. 4-b). Among the four ADs, the reaction rates of the four steps for Con-AD were the lowest while those for MM-AD were the highest, which were consistent with their methane production and enzyme activities (Fig. 1 and Fig. 4-a). This finding indicated that the three methods (MA, MEC, and MM) could accelerate the four AD steps and increase their reaction rates [15,21,32]. The reaction rates of MEC-AD (1.12–1.66 g COD/L/d) were higher than those of MA-AD (1.09–1.26 g COD/L/d), indicating that the enhancing effects of MEC were higher than those of MA. Meanwhile, MM-AD had the highest reaction rates (1.14–1.70 g COD/L/d) for the four AD steps, indicating that MA and MEC had a synergistic effect. The methanogenesis rates of Con-AD and MA-AD were lower than their acidogenesis and acetogenesis rates, whereas the results of MEC-AD and MM-AD were opposite. This finding suggested that MA mainly affects the first three AD steps (hydrolysis, acidogenesis, and acetogenesis) [21] while MEC affects all four AD steps [15]. Additionally, although hydrolysis was enhanced by all three methods, the increase was the lowest among the four AD steps, indicating that it is still the rate-limiting step of AD [44].

3.5. Microbial community analysis

At the end of the stable period (102 d), the digestates from the four ADs were sampled to analyze the microbial communities. The alpha diversity estimators of the digestates from the four ADs are summarized in Table 4. The Chao and Shannon indices were higher for bacteria than archaea, indicating that the abundance and diversity of bacteria were higher than those of archaea [49]. The Chao and Shannon indices of bacteria were lowest for Con-AD among the four ADs, while those of archaea were the highest for Con-AD. These results suggested that these three methods increased the richness and diversity of bacteria and decreased the richness and diversity of archaea. MA can create a suitable environment for facultative bacteria and thereby increase the bacterial richness and diversity and inhibit strict anaerobic methanogens because micro-oxygen is harmful to such organisms [50]. In addition, MEC can enrich both functional bacteria and hydrogenotrophic methanogens [15].

3.5.1. Bacterial community

At the phylum level, Firmicutes, Bacteroidetes, Proteobacteria, Actinobacteriota, and Chloroflexi represent the main hydrolytic and acidogenic bacteria in AD [51]. In this study, their total abundances ranged from 79.05 % to 93.54 % in the four ADs, with the highest observed in MA-AD and the lowest observed in Con-AD (Fig. 5-a). The results indicated that MEC and MA enriched the relative abundances of hydrolytic and acidogenic bacteria [51], with MA exhibiting greater enrichment than MEC. Meanwhile, synergistic effects were observed between MEC and MA. The enrichment of hydrolytic bacteria was consistent with the changes in the hydrolysis rate (Fig. 4-b).

At the genus level, *Clostridium_sensu_stricto_1* was the dominant bacterial genus in the four ADs and had a relative abundance of 29.02–33.19 % (Fig. 5-b). *Romboutsia* was the second bacterial genus (relative abundance of 7.71–10.33 %) in MEC-AD, MA-AD, and MM-AD, while *unclassified_o_Cloacimonadales* ranked second in Con-AD (relative abundance of 11.17 %). Both *Clostridium_sensu_stricto_1* and *Romboutsia* belong to the phylum Firmicutes, which can convert organic matter (such as proteins and carbohydrates) into VFAs and hydrogen [52,53]. Previous studies also have found that *Clostridium_sensu_stricto_1* was the dominant genus in the AD of SM [53,54]. These results indicated that the three methods enhanced hydrolysis and acidogenesis.

The relative abundances of sulfate-reducing bacteria (SRB) and sulfur-oxidizing bacteria (SOB) in the four ADs ranged from 1.06 to 3.84 % and 1.59 to 3.15 %, respectively (Fig. 5-c and d). *Smithella* (belonging

Table 4
The Chao and Shannon indices of bacteria and archaea.

Item		Con-AD	MA-AD	MEC-AD	MM-AD
Bacteria	Chao	743.44	1088.85	786.63	1058.40
	Shannon	3.36	3.81	3.51	3.79
Archaea	Chao	261.22	227.06	226.44	168.91
	Shannon	2.73	1.92	2.38	1.57

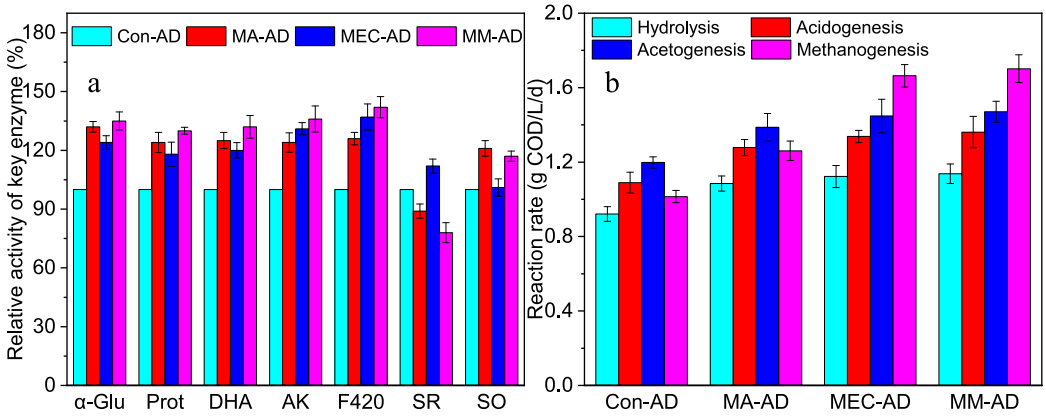


Fig. 4. Reaction rate of four AD steps during the stable stage.

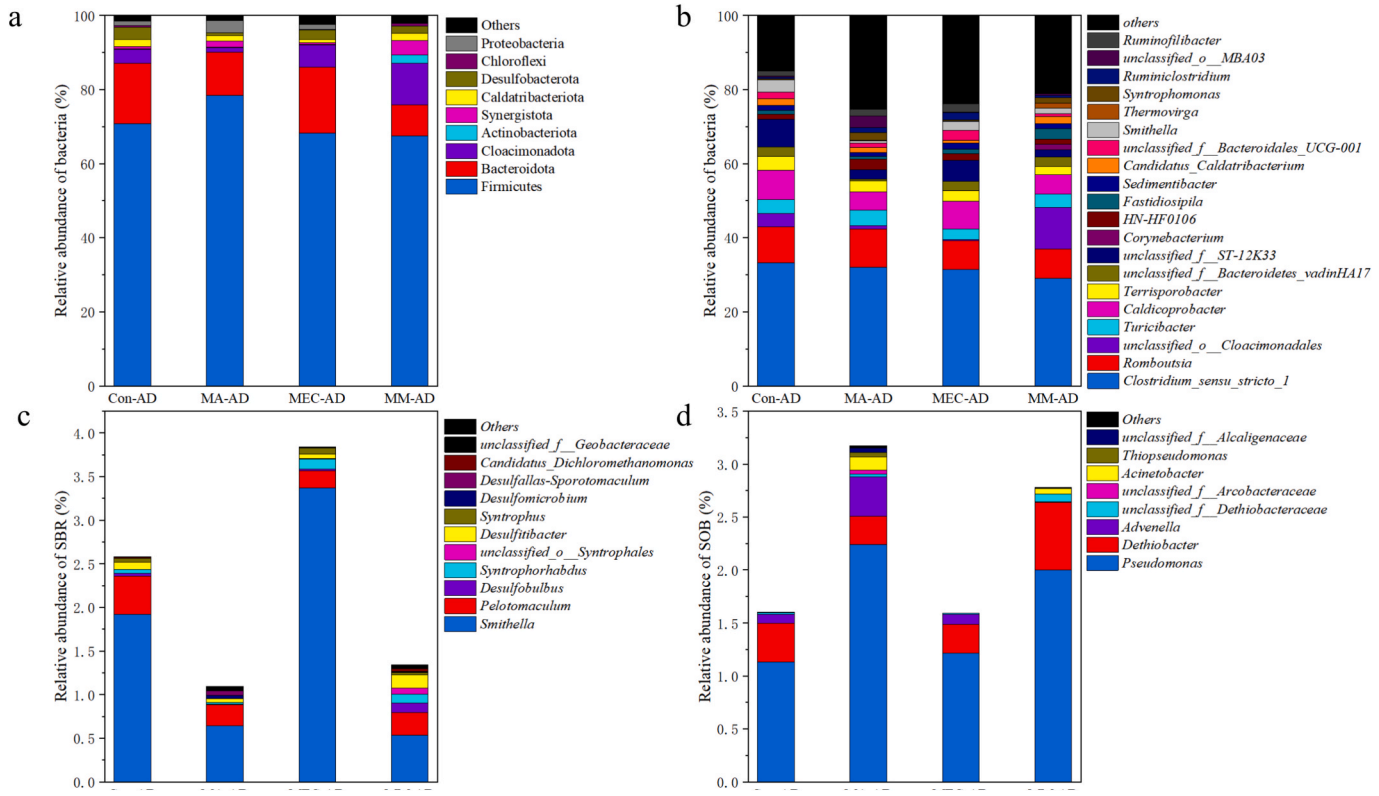


Fig. 5. The bacterial communities in the four ADs during the stable stage.

to Desulfobacterota) was the dominant SRB, with a relative abundance of 0.53–1.92 %, and *Pseudomonas* (belonging to Proteobacteria) was the dominant SOB, with a relative abundance of 1.13–2.24 % [55,56]. The results showed that MEC increased the relative abundance of SRB (from 2.58 % to 3.84 %) but had little effect on SOB, whereas MA decreased the relative abundance of SRB (from 2.58 % to 1.06 %) but increased the relative abundance of SOB (from 1.60 % to 3.15 %). When MEC and MA were co-integrated, the relative abundance of SRB decreased while that of SOB increased (Fig. 5-d). The PICRUSt prediction results also showed that the relative abundance of SRB (including assimilatory and dissimilatory reduction) decreased in MA-AD and MM-AD but slightly increased in MEC-AD compared to that in Con-AD (Table S5). These results indicated that MEC enhanced the anaerobic conversion of sulfur-containing compounds [57,58], while MA enhanced the oxidation of sulfide [51,59]. The changes in SRB and SOB abundances in the four ADs were consistent with the conversion of sulfur (Fig. 3-b).

Correlations among the genus level community composition, main environmental factors, and biogas production were analyzed using RDA

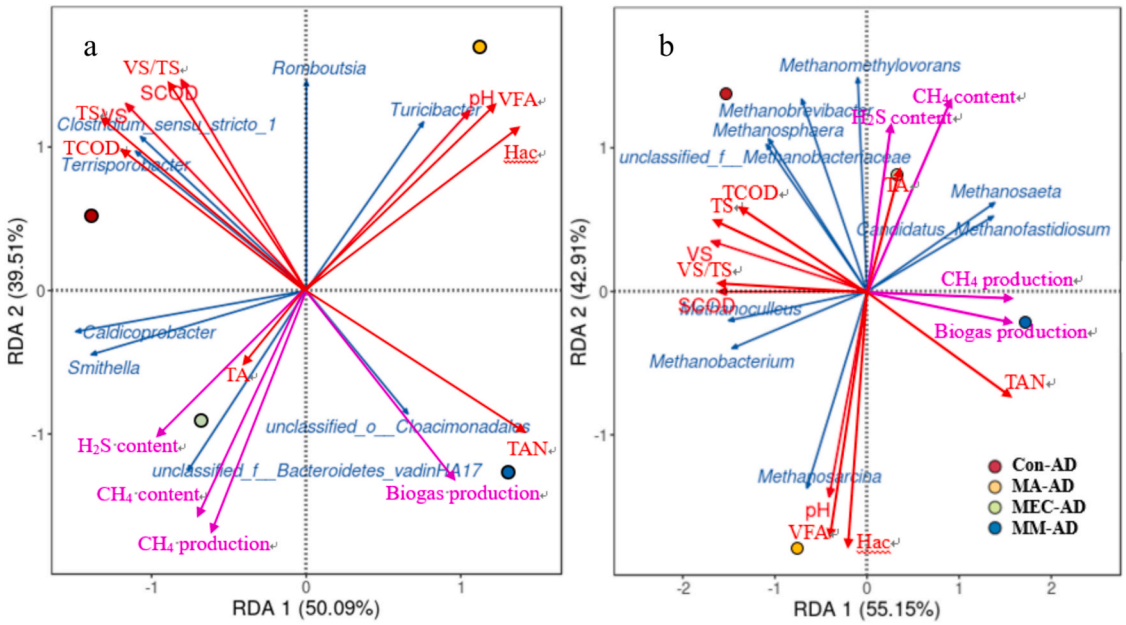


Fig. 6. Redundancy analysis of microbial communities, main environmental factors, and biogas production.

(Fig. 6-a). The total explanation rate was 89.60 %, with the first axis accounting for 50.09 % and the second axis accounting for 39.51 %. The results showed that the dominant bacterial genera (*Clostridium_sensu_stricto_*1 and *Romboutsia*) exhibited highly significant correlations ($p < 0.05$) with the organic characteristics of digestates, including TCOD, SCOD, VS, TS, and VS/TS ratio. Moreover, *Smithella* was significantly correlated with the H_2S concentration in biogas (Fig. 6-a) [60]. The acidic characteristics of the digestates (pH, VFA, and acetic acid) also affected the two dominant bacterial genera.

3.5.2. Archaeal community

The top two predominant archaeal genera differed among the four ADs: *Methanobacterium* (28.10 %) and *Methanosphaera* (22.88 %) were dominant in Con-AD; *Methanosaeta* (37.42 %) and *Methanosphaera* (17.75 %) were dominant in MEC-AD; *Methanosarcina* (51.27 %) and *Methanobacterium* (31.61 %) were dominant in MA-AD; and *Methanosaeta* (53.08 %) and *Candidatus_Methanofastidiosum* (29.56 %) were dominant in MM-AD (Fig. 7). The results showed that MEC enriched *Methanosaeta* and decreased *Methanobacterium*, MA enriched *Methanosarcina* and decreased *Methanosphaera*, and MM enriched *Methanosaeta* and *Candidatus_Methanofastidiosum* and decreased *Methanobacterium* and *Methanosphaera*. Previous studies have reported that *Methanobacterium* and *Methanosphaera* are hydrogenotrophic methanogens, *Methanosaeta* is a strictly acetoclastic methanogen, and *Methanosarcina* and *Candidatus_Methanofastidiosum* are versatile methanogens that can produce methane from multiple substrates [55,61,62]. The results suggested that the predominant methanogenesis pathways in four ADs during the stable stage differed, with hydrogenotrophic pathway observed in Con-AD and acetoclastic and hydrogenotrophic pathways observed in the other three ADs (Table S6) [55,61,62]. It is reported that *Methanosarcina* has a higher growth rate and is more tolerant against different stressors and sudden environmental changes than the other methanogens. *Methanosarcina* can use both the acetoclastic and the hydrogenotrophic pathways [63]. Therefore, versatile methanogenesis can improve the biogas and CH_4 production and the changes in the predominant methanogenesis pathway may explain the increase in methane production by MEC-AD, MA-AD, and MM-AD (Fig. 1-c) [15,62].

RDA (Fig. 6-b) indicated that hydrogenotrophic methanogens were significantly correlated with the organic characteristics of the digestates (TCOD, SCOD, VS, TS, VS/TS), versatile methanogens (*Methanosarcina*) were significantly correlated with the acidic characteristics of the digestates (pH, VFA, and acetic acid), and acetoclastic methanogens (*Methanosaeta*) were significantly correlated with the ammonic characteristics of the digestates (TA and TAN) [60]. *Methanosaeta* and *Candidatus_Methanofastidiosum* were also positively correlated with gas production (biogas and methane production and methane content).

Previous studies have shown that *Methanobacterium* and *Methanosaeta* are electroactive archaea that can perform the DIET and use carbon dioxide to produce methane by obtaining electrons from bacteria such as *Clostridium_sensu_stricto_1* and *Romboutsia* [64,65]. The higher methane content in biogas produced from MEC-AD and MM-AD (Fig. 1-c) indicated that DIET occurred in the two ADs [65–67]. Additionally, the high concentration of acetic acid in the substrate (SM) (Table S4) resulted in the high relative abundance of *Methanosaeta* [68]. Although most methanogens are strict anaerobes and oxygen is lethal to them, some methanogens (e.g., *Methanosarcina* and *Methanocella*) can survive under microaerobic conditions [23]. This may explain why *Methanosarcina* was enriched and dominant in MA-AD [21].

3.6. Implications

In this study, MEC, MA, and their integration (MM) were applied to AD of SM. The results demonstrated that both MEC and MA can enhance the AD of SM and revealed a synergistic effect between them. During the stable stage, the methane yield of SM in MM-AD was the highest (279.94 ± 17.59 mL/g VS), which was 67.70 ± 6.13 % higher than that in Con-AD (Table 2 and Table S2). Meanwhile, the H_2S concentration in the biogas produced in MM-AD (1112 ± 219 ppmV) was 12.94 ± 17.81 % lower than that in Con-AD (1321 ± 342 ppmV). These results indicated that MEC operated in combination with MA can simultaneously increase the energy production of the AD of SM and improve its quality.

A key finding of this study is that MM can significantly increase biogas and methane production and upgrade biogas quality, indicating that more energy can be recovered from SM. This enhancement method is promising and easy to apply in practice because the energy inputs required for MEC and MA are very low [15,20,21]. Although the increase in methane yield was only 113.02 ± 11.58 mL/g VS (from 166.93

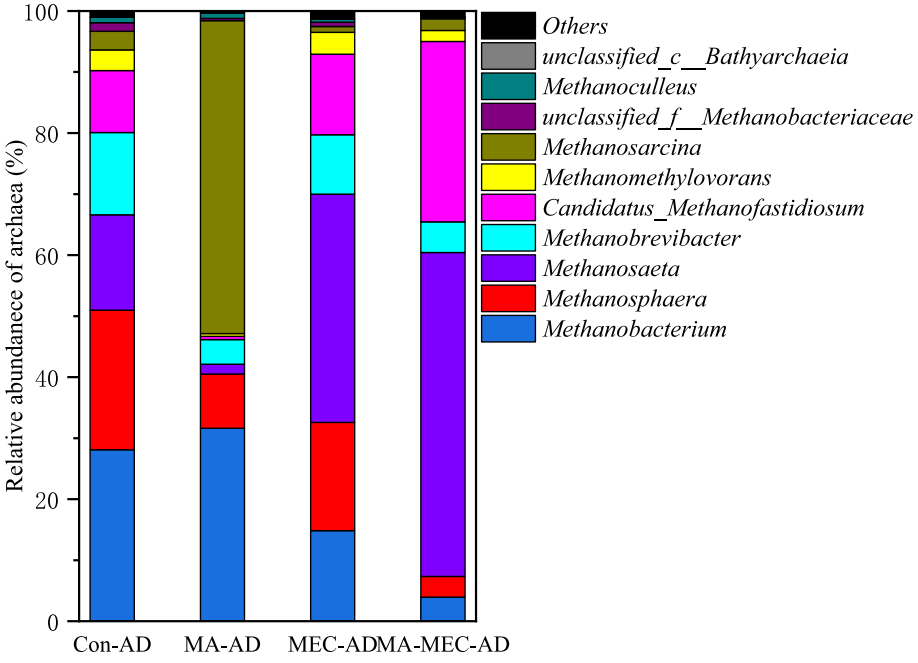


Fig. 7. The archaeal communities at genus level in the four ADs during the stable stage.

± 9.52 mL/g VS to 279.94 ± 17.59 mL/g VS) during the experiment (Table 2) when MM was applied, it could potentially yield 55.33 ± 5.67 billion m^3 if MM is applied to enhance the AD of all SM in China (0.6 billion tons) [2,3]. This significant increase could help alleviate the problem of energy shortages in villages and small-towns in China. Therefore, the results of this study have significant implications for meeting the energy demands of villages and small towns. One of the limitations of this study is that only laboratory-scale experiments were performed. Therefore, further research is needed to apply the results of this study for the practical AD of SM. Additionally, further studies are needed to assess the scalability of the research findings and investigate the long-term sustainability of a scaled-up system, which could help address energy challenges in Chinese villages and small towns.

4. Conclusion

Both MEC and MA enhanced methane production and organic matter removal in the AD of SM, with MEC showing a greater enhancement effect and MEC and MA showing a slight synergistic effect. MEC increased the H_2S concentration in the biogas by enhancing the activity of sulfite reductase, whereas MA and MM decreased it by enhancing the activity of sulfite oxidase. These three methods enriched the hydrolytic and acidogenic bacteria and shifted the predominant methanogenesis pathway. However, further researches on the scalability of the findings and long-term performance and sustainability of scaled-up systems are required to meet the energy demands of villages and small towns in China.

CRediT authorship contribution statement

Yanqiang Di: Writing – original draft, Investigation, Data curation. **Wenxue Hou:** Writing – review & editing, Methodology, Investigation. **Chunlin Bian:** Investigation, Formal analysis. **Tianlong Zheng:** Visualization, Methodology. **Benyi Xiao:** Writing – original draft, Supervision, Resources, Funding acquisition. **Lin Li:** Supervision, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Benyi Xiao reports financial support was provided by the Key Research and Development Program of Inner Mongolia Autonomous Region, China.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2025.163319.

Data availability

The data that has been used is confidential.

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